Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | WARM UPS

Name: _____ Date: _____

1 | Multiple Choice NO CALCULATOR

Recent Review | 3.7 Implicit Differentiation

For the curve $x^2 + 2xy + 4y^2 = 12$, which pair of points has horizontal tangent lines?

A. $(4, -1)$ and $(-4, 1)$

B. $(0, \sqrt{3})$ and $(0, -\sqrt{3})$

C. $(2, -2)$ and $(-2, 2)$

D. $(2, 1)$ and $(-2, -1)$

E. No horizontal tangent lines

Show your reasoning here.

2 | Multiple Choice CALCULATOR PERMITTED

Spiral Review | Product Rule + Chain Rule + Trigonometry

Let $f(x) = (x^2 + 1)\cos(0.7x)$. Which equation best approximates the line tangent to $y = f(x)$ at $x = 1.4$? Use radians; round coefficients to three decimals.

A. $y - 1.649 = 0.161(x - 1.4)$

B. $y - 0.161 = -1.649(x - 1.4)$

C. $y - 1.649 = 1.649(x - 1.4)$

D. $y - 1.649 = -0.161(x - 1.4)$

E. $y - 1.649 = -0.161(x + 1.4)$

Show your reasoning here.

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | PROJECTABLE EXERCISES

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q01

Derivatives of Inverse Trigonometric Functions

1 Find the derivative of

$h(x) = x\sec^{-1}(3x)$

Exercise 1 of 13 | Same original question

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q02

Derivatives of Inverse Trigonometric Functions

2 Find the derivative of

$f(x) = x\sqrt{1-x^2} + \cos^{-1}(x^3)$

Exercise 2 of 13 | Same original question

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | PROJECTABLE EXERCISES

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q03

Derivatives of Inverse Trigonometric Functions

3 Find an equation for the tangent line to the graph of

$y = \tan^{-1} x \quad \text{at} \quad (1, \pi/4)$

Exercise 3 of 13 | Same original question

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q04

Derivatives of Inverse Trigonometric Functions

4 Find the inverse-function derivative

$(f^{-1})'(2) \quad \text{if} \quad f(x) = x^3 + 2x - 1$

Exercise 4 of 13 | Same original question

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | PROJECTABLE EXERCISES

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q05

Derivatives of Inverse Trigonometric Functions

5 Find the inverse-function derivative

$$(f^{-1})'(6) \quad \text{if} \quad f(x) = x^3 - \frac{2}{3}$$

Exercise 5 of 13 | Same original question

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q06

Derivatives of Inverse Trigonometric Functions

6 Find the derivative of y with respect to the appropriate variable.

$$y = \cos^{-1}(x^2)$$

Exercise 6 of 13 | Same original question

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | PROJECTABLE EXERCISES

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q07

Derivatives of Inverse Trigonometric Functions

7 Find the derivative of y with respect to the appropriate variable.

$$y = \sin^{-1}\left(\frac{3}{t^2}\right)$$

Exercise 7 of 13 | Same original question

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q08

Derivatives of Inverse Trigonometric Functions

8 A particle moves along the x-axis, with t ≥ 0. Find its velocity at t = 1.

$$x(t) = \tan^{-1}(t^2), \quad t = 1$$

Exercise 8 of 13 | Same original question

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | PROJECTABLE EXERCISES

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q09

Derivatives of Inverse Trigonometric Functions

9 Find the derivative of y with respect to the appropriate variable.

$y = \sec^{-1}(2s + 1)$

Exercise 9 of 13 | Same original question

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q10

Derivatives of Inverse Trigonometric Functions

10 Find the derivative of y with respect to the appropriate variable.

$y = \csc^{-1}(x^2 + 1), \quad x > 0$

Exercise 10 of 13 | Same original question

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | SECTION 3.8 | PROJECTABLE EXERCISES

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q11

Derivatives of Inverse Trigonometric Functions

11 Find the derivative of y with respect to the appropriate variable.

$y = \sqrt{s^2 - 1} - \sec^{-1}(s)$

Exercise 11 of 13 | Same original question

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q12

Derivatives of Inverse Trigonometric Functions

12 Find an equation for the tangent line to the graph at x = 2.

$y = \sec^{-1} x, \quad x = 2$

Exercise 12 of 13 | Same original question

Derivatives of Inverse Trigonometric Functions

AP CALCULUS AB | PROJECTABLE QUESTION 3.8 | Q13

Derivatives of Inverse Trigonometric Functions

13 Find an equation for the tangent line to the graph at x = 3.

$$y = \sin^{-1}\left(\frac{x}{4}\right), \quad x = 3$$

Exercise 13 of 13 | Same original question